Ali AGHABABAEI-HARANDI

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Google Scholar

Summary

Final-year Ph.D. candidate at Université Grenoble Alpes, supervised by Professor Massih-Reza Amini. Research focuses on model efficiency and representation learning within machine learning and deep learning, with the goal of designing scalable and practical AI systems.

Education

Université Grenoble Alpes *2023 – Present*
Ph.D. in Computer Science

- **Research:** Deep Learning Efficiency, Representation Learning
- Supervised by Professor Massih-Reza Amini, APTIKAL team

Sharif University of Technology *2018 – 2022*
M.Sc. in Computer Science

- **Coursework:** Deep Learning, Speech Processing

University of Tehran *2013 – 2018*
B.Sc. in Electrical Engineering

- **Coursework:** Linear Algebra, Pattern Recognition, Digital Signal Processing

Research Interests

- **Deep Learning Efficiency:** Tensor Decomposition, Knowledge Distillation, Memory-Efficient Training
- **Representation Learning:** Semi-Supervised Learning, Pseudo-Labeling, Contrastive Learning
- **Generative AI:** Probabilistic Modeling, Image Denoising, Image Super-Resolution

Research Experience

Ph.D. Thesis *2023 – Present*
APTIKAL team, Université Grenoble Alpes

- Developing a **memory-efficient** training method by improving greedy supervised learning; proposed preserving information during local training via neural information estimation.
- Extended the **Forward-Forward** algorithm to Vision Transformers by introducing early cross-attention between label tokens and feature representations.
- Investigated semi-supervised learning via **pseudo-labeling** through theoretical analysis of bias stability; proposed an adaptive **Gaussian geometric** loss to shape latent representations, accelerate convergence, and reduce majority-class bias.
- Reduced computational complexity and parameter count of pre-trained models through a **tensor decomposition** approach with a custom **constraint-based** loss function for optimal rank selection while preserving accuracy.

Research Collaborator *2021 – 2022*
University of Basel

- Analysed inverse problems (image denoising, limited-view CT, wave scattering) using invertible neural networks.
- Approximated data distributions with normalizing flows to identify out-of-distribution samples, improving model robustness and reliability.

M.Sc. Thesis *2018 – 2021*

Electronic Research Institute, Sharif University of Technology

- Proposed a patch-wise feature analysis approach for identifying forgery in video frames, enhancing deepfake detection capabilities.
- Developed a method to improve generalization and robustness of deepfake detection models.

B.Sc. Thesis *2016 – 2017*

University of Tehran

- Designed and implemented a no-reference video quality meter assessing blockiness and blurriness distortions.

Publications

JEPAMatch: Geometric Representation Shaping for Semi-Supervised Learning *March 2026*

Ali Aghababaei-Harandi, Aude Sportisse, Massih-Reza Amini

arXiv preprint — [link]

Unified Framework for Pre-trained Neural Network Compression via Decomposition and Optimized Rank Selection *September 2025*

Ali Aghababaei-Harandi, Massih-Reza Amini

ECML-PKDD 2025 — [link]

Deep Variational Inverse Scattering *March 2023*

AmirEhsan Khorashadizadeh, Ali Aghababaei-Harandi, Tin Vlašić, Hieu Nguyen, Ivan Dokmanić

EuCAP 2023 — [link]

Conditional Injective Flows for Bayesian Imaging *February 2023*

AmirEhsan Khorashadizadeh, Konik Kothari, Leonardo Salsi, Ali Aghababaei-Harandi, Maarten de Hoop, Ivan Dokmanić

IEEE Transactions on Computational Imaging — [link]

Talks & Presentations

Efficient Deep Learning *December 2025*

Mathematical Fundamentals of AI Workshop, Sorbonne Center for AI (SCAI), Paris, France

Presentation on Pre-trained Neural Network Compression *September 2025*

European Conference on Machine Learning (ECML-PKDD), Porto, Portugal

Conférence sur l'Apprentissage Automatique (CAp), Dijon, France

Skills

Programming: Python, PyTorch, TensorFlow, JAX, Hugging Face, Git

Theoretical: Pattern Recognition, Design and Analysis of Algorithms, Deep Learning